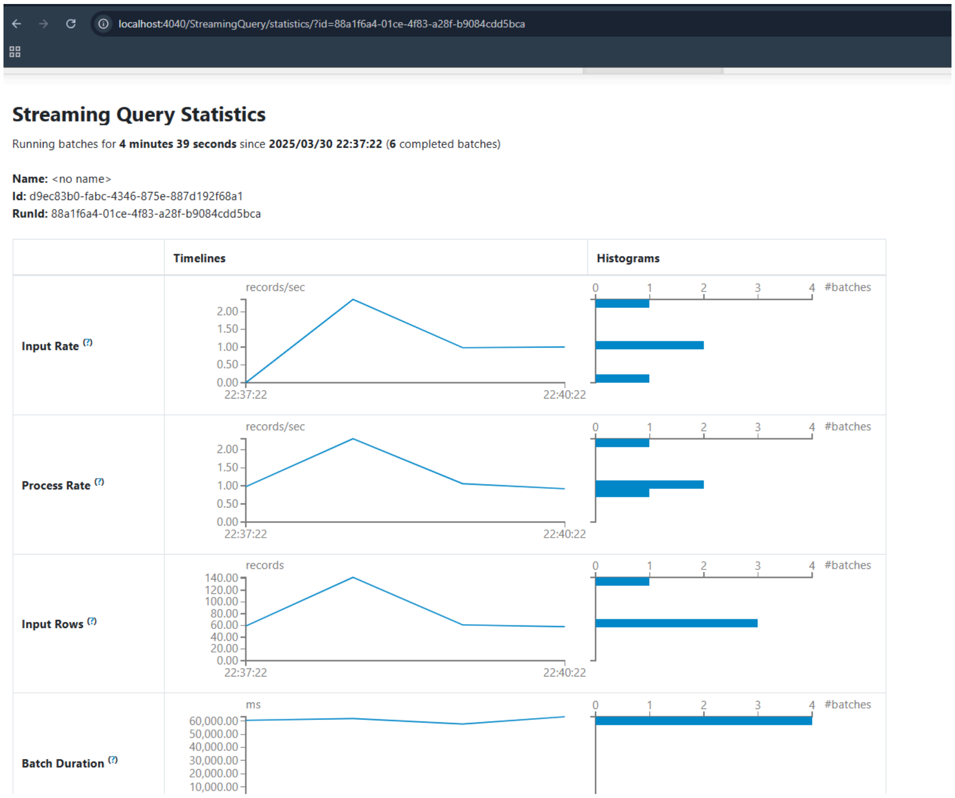
- Generating data through producers
- asf

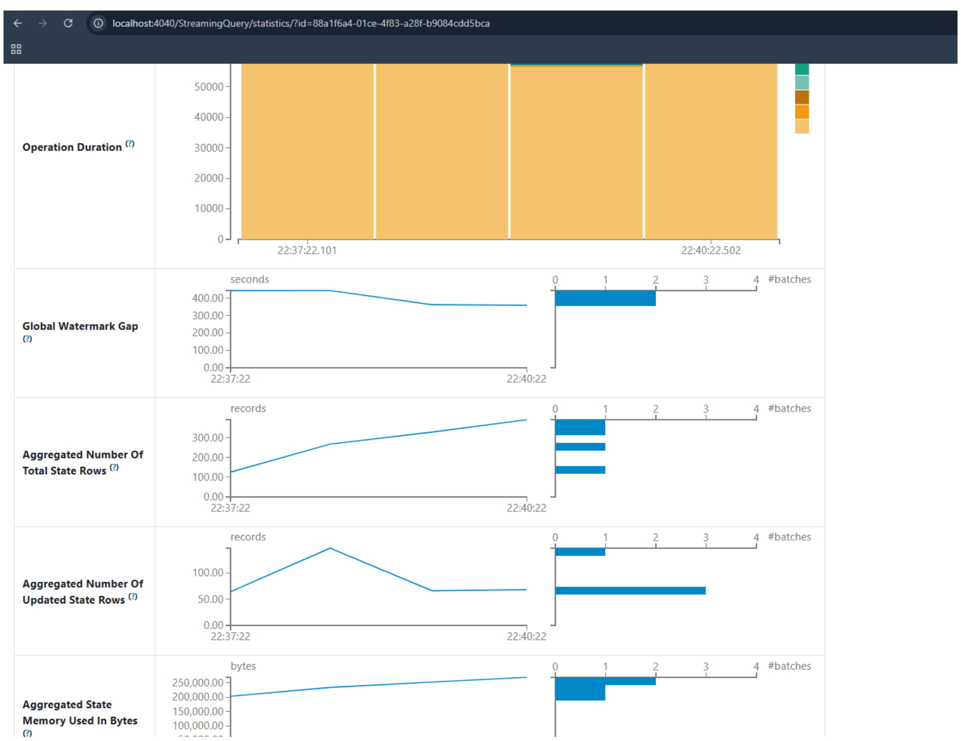
```
Sent: {'sensor_id': 'Sensor_3', 'timestamp': '2025-03-30T22:17:20', 'vehicle_count': 81, 'average_speed': 29.9, 'congestion_level': 'MEDIUM'}
Sent: {'sensor_id': 'Sensor_2', 'timestamp': '2025-03-30T22:17:21', 'vehicle_count': 37, 'average_speed': 58.1, 'congestion_level': 'LOW'}
PS C:\LUMS\Semester 2\Data Engineering\Lab 5\ai601-data-engineering\labs\lab5> python .\traffic_producer.py
Sent: {'sensor_id': 'Sensor_4', 'timestamp': '2025-03-30T22:29:01', 'vehicle_count': 83, 'average_speed': 3.6, 'congestion_level': 'LOW'}
Sent: {'sensor_id': 'Sensor_2', 'timestamp': '2025-03-30T22:29:02', 'vehicle_count': 50, 'average_speed': 63.8, 'congestion_level': 'LOW'}
Sent: {'sensor_id': 'Sensor_3', 'timestamp': '2025-03-30T22:29:03', 'vehicle_count': 59, 'average_speed': 93.1, 'congestion_level': 'HIGH'}
Sent: {'sensor_id': 'Sensor_0', 'timestamp': '2025-03-30T22:29:04', 'vehicle_count': 37, 'average_speed': 71.1, 'congestion_level': 'MEDIUM'}
Sent: {'sensor_id': 'Sensor_4', 'timestamp': '2025-03-30T22:29:05', 'vehicle_count': 5, 'average_speed': 10.0, 'congestion_level': 'MEDIUM'}
Sent: {'sensor_id': 'Sensor_1', 'timestamp': '2025-03-30T22:29:06', 'vehicle_count': 93, 'average_speed': 44.0, 'congestion_level': 'MEDIUM'}
Sent: {'sensor_id': 'Sensor_0', 'timestamp': '2025-03-30T22:29:07', 'vehicle_count': 95, 'average_speed': 119.1, 'congestion_level': 'MEDIUM'}
Sent: {'sensor_id': 'Sensor_3', 'timestamp': '2025-03-30T22:29:08', 'vehicle_count': 34, 'average_speed': 59.0, 'congestion_level': 'LOW'}
Sent: {'sensor_id': 'Sensor_3', 'timestamp': '2025-03-30T22:29:09', 'vehicle_count': 64, 'average_speed': 44.4, 'congestion_level': 'HIGH'}
Sent: {'sensor_id': 'Sensor_2', 'timestamp': '2025-03-30T22:29:10', 'vehicle_count': 76, 'average_speed': 110.4, 'congestion_level': 'MEDIUM'}
Sent: {'sensor_id': 'Sensor_4', 'timestamp': '2025-03-30T22:29:11', 'vehicle_count': 10, 'average_speed': 73.3, 'congestion_level': 'MEDIUM'}
Sent: {'sensor_id': 'Sensor_1', 'timestamp': '2025-03-30T22:29:12', 'vehicle_count': 39, 'average_speed': 88.7, 'congestion_level': 'LOW'}
Sent: {'sensor_id': 'Sensor_4', 'timestamp': '2025-03-30T22:29:13', 'vehicle_count': 12, 'average_speed': 102.0, 'congestion_level': 'LOW'}
Sent: {'sensor_id': 'Sensor_1', 'timestamp': '2025-03-30T22:29:14', 'vehicle_count': 2, 'average_speed': 36.9, 'congestion_level': 'MEDIUM'}
Sent: {'sensor_id': 'Sensor_3', 'timestamp': '2025-03-30T22:29:15', 'vehicle_count': 79, 'average_speed': 48.5, 'congestion_level': 'LOW'}
Sent: {'sensor_id': 'Sensor_1', 'timestamp': '2025-03-30T22:29:16', 'vehicle_count': 77, 'average_speed': 18.3, 'congestion_level': 'MEDIUM'}

Ln 32, Col 26   Spaces: 4   UTF-8   CRLF   {} Python
```

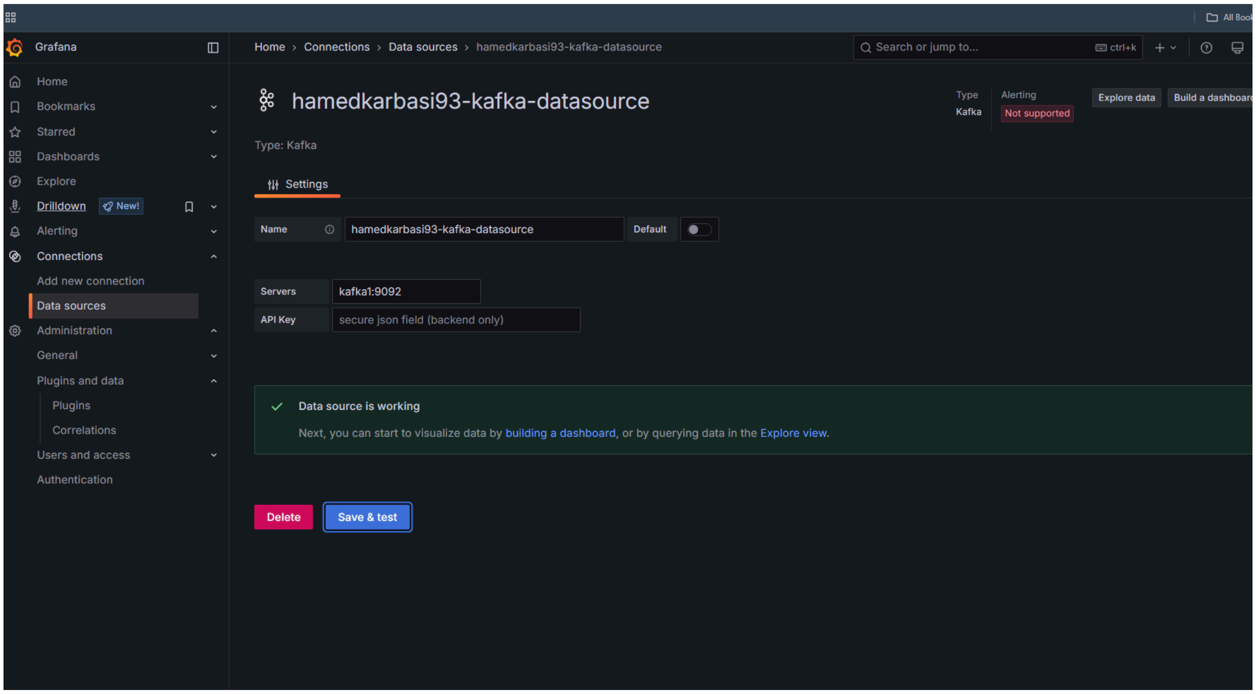
Streaming

Below SS are taken from Spark Built in UI


localhost:4040/StreamingQuery/statistics/?id=88a1f6a4-01ce-4f83-a28f-b9084cdd5bca
Streaming Query Statistics
Running batches for 4 minutes 39 seconds since 2025/03/30 22:37:22 (6 completed batches)
Name: <no name>
Id: d9ec83b0-fabc-4346-875e-887d192f68a1
RunId: 88a1f6a4-01ce-4f83-a28f-b9084cdd5bca
Timelines
Histograms
Input Rate
Process Rate
Input Rows
Batch Duration
records/sec
records
ms
#batches
22:37:22
22:40:22


localhost:4040/StreamingQuery/statistics/?id=88a1f6a4-01ce-4f83-a28f-b9084cdd5bca
Operation Duration
22:37:22.101
22:40:22.502
Global Watermark Gap
seconds
Aggregated Number Of Total State Rows
records
Aggregated Number Of Updated State Rows
Aggregated State Memory Used In Bytes
bytes
#batches
22:37:22
22:40:22

Finally grafana configure -> took me the most time



Data is being picked up by grafana.